

ImmunoStruct enables multimodal deep learning for immunogenicity prediction

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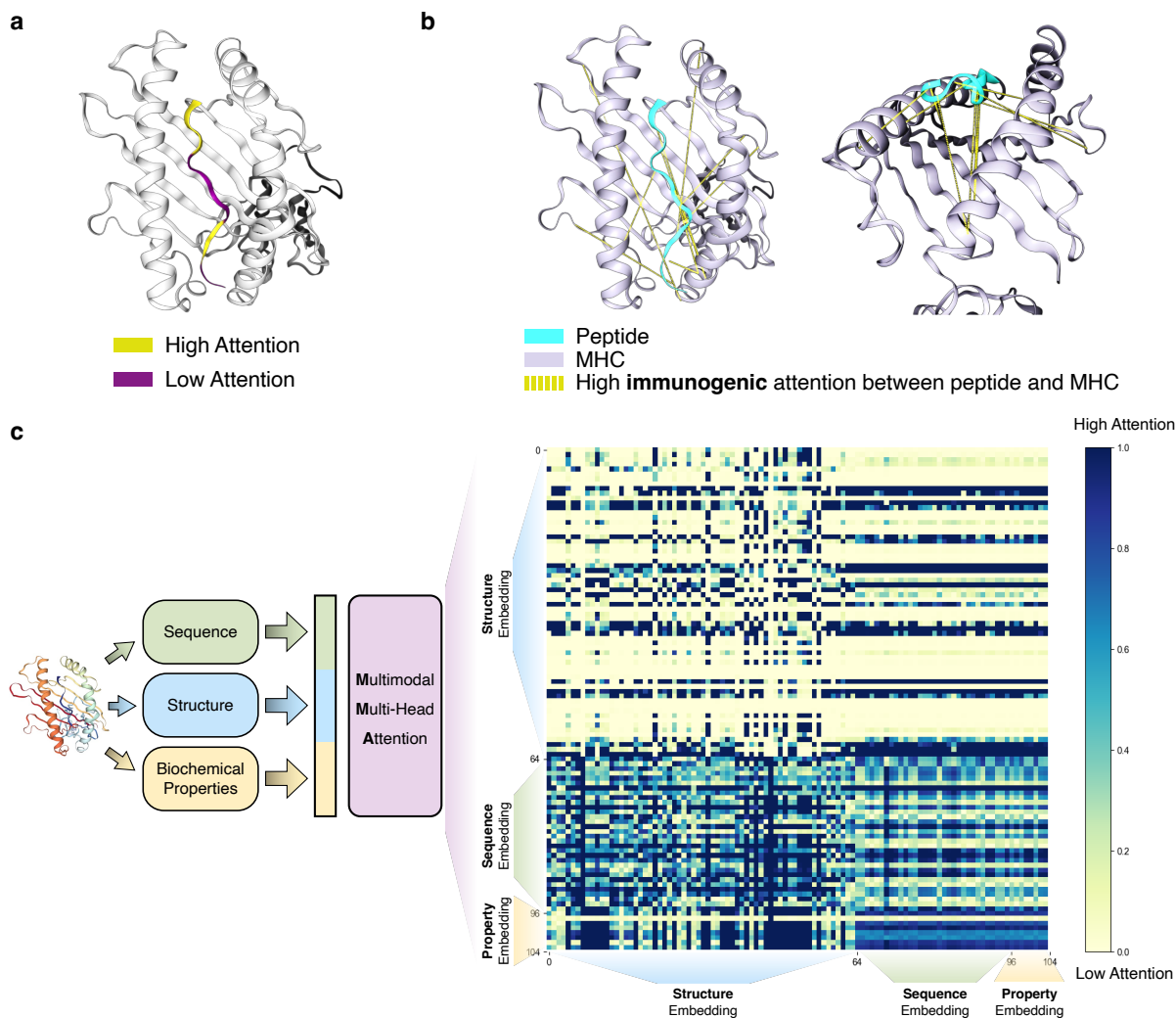


Figure S1: **ImmunoStruct attention visualization examples on the peptide-MHC structure, and for the multi-modal embeddings.** (a) The EGNN module attention layer can be extracted and visualized for both high (yellow) and low (purple) attention weights, directly on the peptide docked into the MHC protein. (b) The EGNN node-to-node attention weights between the peptide and the MHC protein can also be visualized as dotted lines (yellow) between the docked peptide and MHC protein. (c) The MMA learned attention matrix visualized as a heatmap across embeddings generated from the sequence, structure, and biochemical property data modalities.

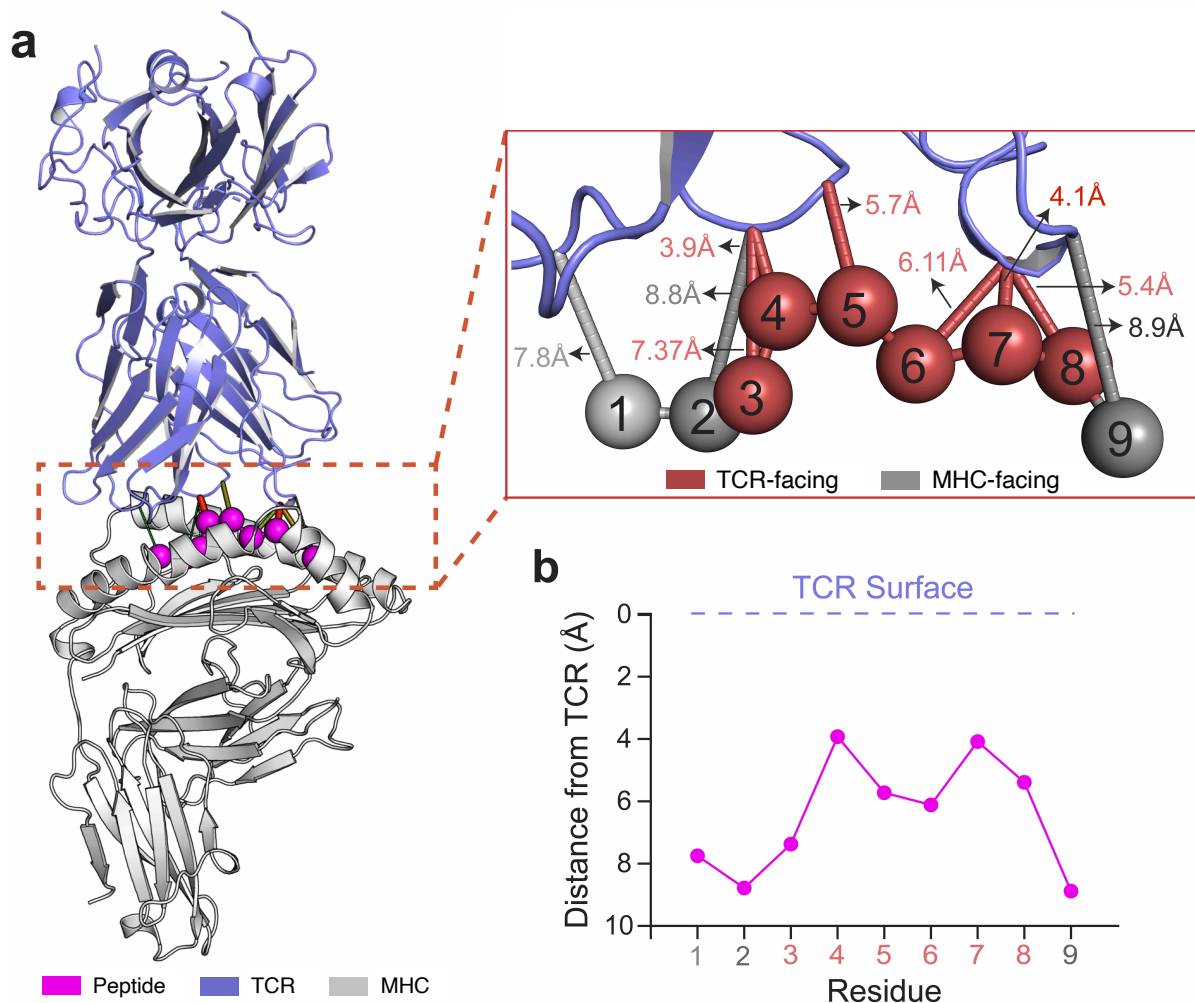


Figure S2: **Quantitative peptide-TCR contact analysis on a canonical TCR-peptide-MHC crystal complex (Tax-A6-HLA-A2).** (a) Structural overlay of the A6 TCR bound to HLA-A2 presenting the highly immunogenic Tax peptide (magenta, PDB: 1BD2). The MHC heavy chain is shown in gray, the TCR in blue, and the peptide as C α spheres connected by sticks. Dashed lines illustrate peptide-C α to TCR-C α distances, listed in Å. (b) Quantification of peptide-TCR separation. For each peptide position (Residues 1-9), the minimal C α -C α distance to the TCR was measured and plotted. The immunogenic Tax peptide (magenta) shows reduced separation at central positions, consistent with a bulged conformation.

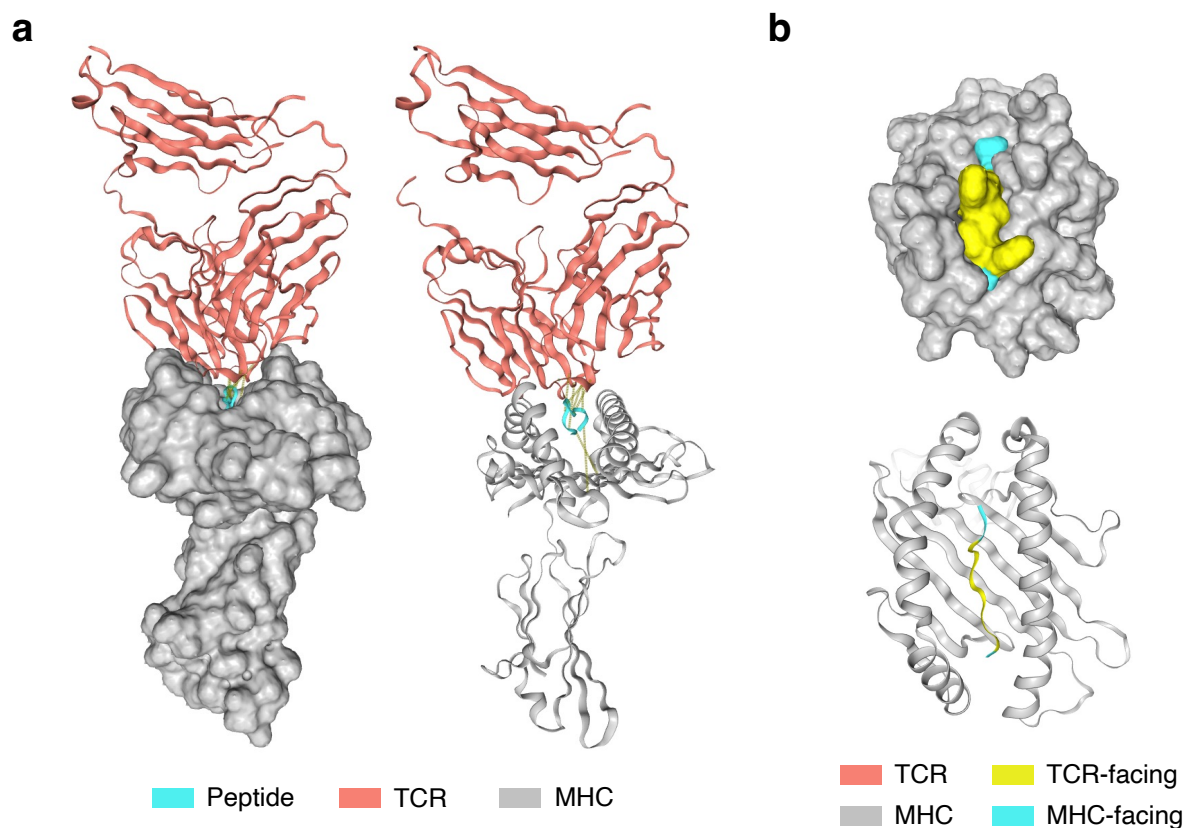


Figure S3: **Additional structural illustrations of the MHC-facing and TCR-facing positions of the peptide.** (a) Surface view (left) and cartoon view (right) of the TCR-peptide-MHC. (b) Zoom-in surface view (top) and cartoon view (bottom) of the peptide-MHC, with TCR-facing and MHC-facing segments of the peptide differently colored.

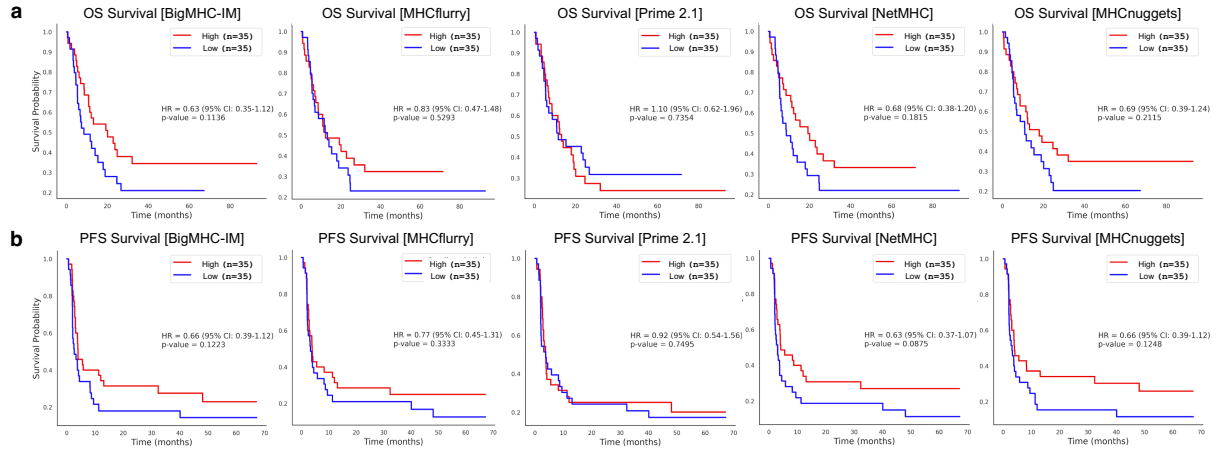


Figure S4: **Kaplan-Meier survival plots for other methods.** Kaplan-Meier survival plots on $n = 70$ cancer patients for BigMHC, MHCflurry, Prime 2.1, NetMHC, and MHCnuggets showing these methods' respective stratification performance on the cancer clinical survival dataset for **(a)** Overall Survival (OS) and **(b)** Progression-free survival (PFS). The median split for predicted immunogenic peptide-MHC count was used, as described in Fig 5.



Figure S5: **Peptides length distributions.** Length distribution of immunogenic vs. non-immunogenic peptides in the train, validation, test sets and the ImmunoStruct predictions. The peptide length distributions between the ImmunoStruct prediction on the test set (4th panel) are similar to the training distributions (1st panel).

Table S1: **Ablation on the model components on the IEDB dataset of infectious diseases.** All experiments were performed under 5 random seeds, and the mean and sample standard deviation (AUROC, AUPRC) or standard error of the mean (Mean PPVn) are displayed. For each metric, the best candidate is bolded. Bchems stands for biochemical properties and MMA stands for multimodal multi-head attention.

Structure	Components			Transfer Learning?	AUROC	AUPRC	mean PPVn
	Sequence	Bchems	MMA				
✓	✓	✓	✓	✓	0.882 (0.005)	0.696 (0.020)	0.514 (0.021)
✓	✓	✓		✓	0.840 (0.006)	0.554 (0.028)	0.441 (0.029)
	✓	✓		✓	0.844 (0.006)	0.561 (0.017)	0.442 (0.024)
✓	✓			✓	0.844 (0.005)	0.569 (0.014)	0.440 (0.035)
	✓			✓	0.845 (0.007)	0.568 (0.024)	0.427 (0.017)
✓				✓	0.805 (0.006)	0.497 (0.022)	0.414 (0.027)
✓	✓	✓	✓		0.860 (0.013)	0.615 (0.045)	0.462 (0.015)
✓	✓	✓			0.840 (0.006)	0.547 (0.018)	0.440 (0.032)
	✓	✓			0.842 (0.006)	0.554 (0.020)	0.419 (0.017)
✓	✓				0.840 (0.007)	0.547 (0.024)	0.444 (0.028)
	✓				0.842 (0.006)	0.553 (0.014)	0.430 (0.009)
✓					0.775 (0.012)	0.442 (0.022)	0.433 (0.020)

Table S2: **Ablation on contrastive learning components on the CEDAR dataset of cancer neoepitopes.** All experiments were performed under 5 random seeds, and the mean and sample standard deviation (AUROC, AUPRC) or standard error of the mean (Mean PPVn) are displayed. For each metric, the best candidate is bolded.

Components		Transfer Learning?	AUROC	AUPRC	mean PPVn
Pairwise Similarity	Feature Independence				
✓	✓	✓	0.771 (0.024)	0.433 (0.069)	0.365 (0.057)
✓		✓	0.765 (0.028)	0.423 (0.075)	0.357 (0.034)
	✓	✓	0.752 (0.033)	0.421 (0.072)	0.366 (0.044)
		✓	0.756 (0.034)	0.426 (0.056)	0.362 (0.046)
✓	✓		0.727 (0.037)	0.405 (0.082)	0.387 (0.049)
✓			0.725 (0.030)	0.398 (0.073)	0.352 (0.063)
	✓		0.709 (0.042)	0.375 (0.079)	0.327 (0.061)
			0.723 (0.027)	0.391 (0.068)	0.358 (0.043)

Table S3: **Comparison of immunogenicity prediction methods on the IEDB dataset of infectious diseases and the CEDAR dataset of cancer neoepitopes.** All experiments were performed under 5 random seeds. The mean values are reported, while the sample standard deviation (AUROC, AUPRC) or standard error of the mean (Mean PPVn) are displayed in parentheses. For each metric, the best candidate is bolded.

Model	IEDB dataset			CEDAR dataset		
	AUROC	AUPRC	mean PPVn	AUROC	AUPRC	mean PPVn
Prime-2.1 [22]	0.538 (0.012)	0.212 (0.031)	0.207 (0.013)	0.645 (0.026)	0.259 (0.023)	0.295 (0.013)
NetMHCpan [13]	0.537 (0.027)	0.214 (0.020)	0.220 (0.008)	0.559 (0.049)	0.210 (0.037)	0.211 (0.020)
MHCnuggets [14]	0.546 (0.023)	0.246 (0.019)	0.233 (0.014)	0.530 (0.014)	0.210 (0.017)	0.175 (0.010)
MHCflurry [15]	0.577 (0.021)	0.260 (0.026)	0.273 (0.013)	0.658 (0.015)	0.304 (0.032)	0.329 (0.013)
DeepNeo [23]	0.767 (0.032)	0.438 (0.024)	0.411 (0.008)	0.535 (0.016)	0.261 (0.053)	0.222 (0.031)
BigMHC-EL [16]	0.588 (0.018)	0.269 (0.016)	0.290 (0.011)	0.632 (0.034)	0.248 (0.039)	0.245 (0.019)
BigMHC-IM [16]	0.684 (0.028)	0.319 (0.048)	0.333 (0.015)	0.771 (0.040)	0.373 (0.062)	0.357 (0.016)
BigMHC _{retrained} [16]	0.793 (0.013)	0.462 (0.027)	0.445 (0.004)	0.682 (0.012)	0.310 (0.020)	0.325 (0.013)
NeoaPred [18]	—	—	—	0.556 (0.016)	0.267 (0.022)	0.292 (0.036)
ImmunoStruct (ours)	0.882 (0.005)	0.696 (0.020)	0.514 (0.009)	0.771 (0.024)	0.433 (0.069)	0.365 (0.057)

Table S4: **Comparison of immunogenicity prediction on the IEDB dataset of infectious diseases from training over common alleles and testing on only the lowest frequency alleles.** Only peptide-MHC samples corresponding to the 10 least represented alleles were used in the test set for the rare allele model.

Model	AUROC	AUPRC	mean PPVn
ImmunoStruct	0.882	0.696	0.514
ImmunoStruct _{rare alleles}	0.610	0.775	0.720

Table S5: **IEDB and CEDAR dataset statistics by allele.** Counts are per unique peptide-allele pair.

HLA allele	IEDB dataset			CEDAR dataset		
	Total <i>n</i>	Immunogenic <i>n</i> (%)	Non-immunogenic <i>n</i> (%)	Total <i>n</i>	Immunogenic <i>n</i> (%)	Non-immunogenic <i>n</i> (%)
HLA-A*01:01	1276	281 (22.0%)	995 (78.0%)	38	6 (15.8%)	32 (84.2%)
HLA-A*02:01	3956	1895 (47.9%)	2061 (52.1%)	1168	311 (26.6%)	857 (73.4%)
HLA-A*02:03	29	28 (96.6%)	1 (3.4%)	9	4 (44.4%)	5 (55.6%)
HLA-A*02:06	44	36 (81.8%)	8 (18.2%)	24	7 (29.2%)	17 (70.8%)
HLA-A*03:01	2140	227 (10.6%)	1913 (89.4%)	158	23 (14.6%)	135 (85.4%)
HLA-A*11:01	2435	245 (10.1%)	2190 (89.9%)	252	30 (11.9%)	222 (88.1%)
HLA-A*23:01	1212	21 (1.7%)	1191 (98.3%)	—	—	—
HLA-A*24:02	1968	390 (19.8%)	1578 (80.2%)	294	28 (9.5%)	266 (90.5%)
HLA-A*26:01	692	22 (3.2%)	670 (96.8%)	8	2 (25.0%)	6 (75.0%)
HLA-A*30:01	32	16 (50.0%)	16 (50.0%)	32	4 (12.5%)	28 (87.5%)
HLA-A*30:02	855	26 (3.0%)	829 (97.0%)	4	4 (100.0%)	0 (0.0%)
HLA-A*31:01	28	19 (67.9%)	9 (32.1%)	17	11 (64.7%)	6 (35.3%)
HLA-A*32:01	24	9 (37.5%)	15 (62.5%)	—	—	—
HLA-A*33:01	1605	23 (1.4%)	1582 (98.6%)	4	1 (25.0%)	3 (75.0%)
HLA-A*68:01	1425	43 (3.0%)	1382 (97.0%)	17	4 (23.5%)	13 (76.5%)
HLA-A*68:02	62	29 (46.8%)	33 (53.2%)	17	1 (5.9%)	16 (94.1%)
HLA-B*07:02	1263	411 (32.5%)	852 (67.5%)	306	20 (6.5%)	286 (93.5%)
HLA-B*08:01	322	215 (66.8%)	107 (33.2%)	39	8 (20.5%)	31 (79.5%)
HLA-B*15:01	168	130 (77.4%)	38 (22.6%)	149	17 (11.4%)	132 (88.6%)
HLA-B*35:01	403	238 (59.1%)	165 (40.9%)	109	15 (13.8%)	94 (86.2%)
HLA-B*40:01	1209	89 (7.4%)	1120 (92.6%)	17	9 (52.93%)	8 (47.1%)
HLA-B*44:02	971	55 (5.7%)	916 (94.3%)	13	8 (61.5%)	5 (38.5%)
HLA-B*44:03	57	41 (71.9%)	16 (28.1%)	17	3 (17.6%)	14 (82.4%)
HLA-B*51:01	529	61 (11.5%)	468 (88.5%)	7	2 (28.6%)	5 (71.4%)
HLA-B*53:01	359	40 (11.1%)	319 (88.9%)	—	—	—
HLA-B*57:01	110	95 (86.4%)	15 (13.6%)	11	2 (18.2%)	9 (81.8%)
HLA-B*58:01	165	71 (43.0%)	94 (57.0%)	—	—	—
All	23,339	4,756 (20.4%)	18,583 (79.6%)	2,710	520 (19.2%)	2,190 (80.8%)